

The Genetic Architecture of Wisdom

A Genome-Wide Association Study in 131,870 Individuals

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UC San Diego



All of Us
RESEARCH PROGRAM

What is Wisdom?

Historically:

- Discussed for millennia across cultures: ancient Egyptian texts, Hindu scripture, Buddhist and Confucian philosophy, Greek thinkers, Judeo-Christian tradition
- Hall (1922) first theorized wisdom as a function of aging: calmness, impartiality, moral knowledge from experience

What is Wisdom?

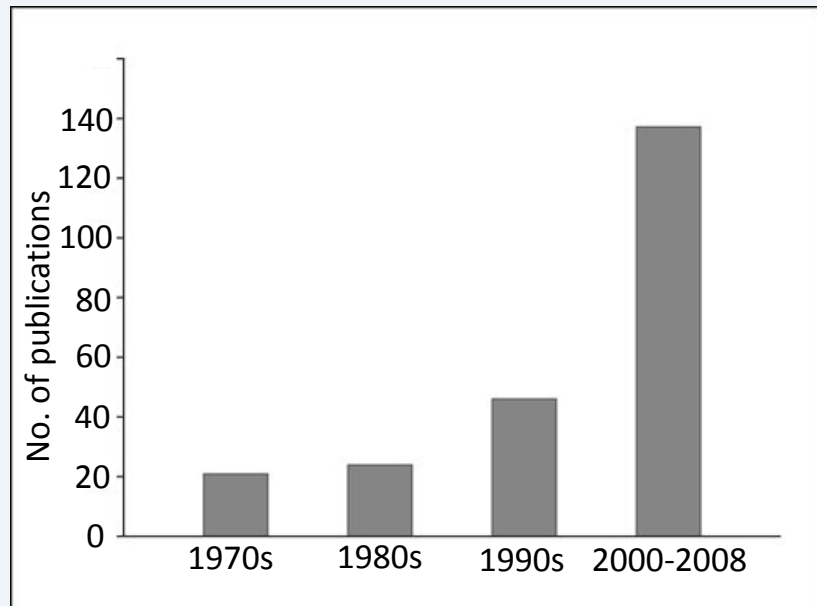
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**How would
you define
wisdom?**

Quantifying the Modern Science of Wisdom

Growth of Empirical Wisdom Research (1970–2017)



Jeste DV, Lee EE, 2019. The Emerging Empirical Science of Wisdom: Definition, Measurement, Neurobiology, Longevity, and Interventions.

Modern definition: SD-WISE-7 Scale

7 Core Components:

- Emotional Regulation
- Prosocial Behaviors
- Self-Reflection
- Divergent Perspectives
- Decisiveness
- Social Advising
- Spirituality

Thomas et al., 2019

Genetic Mapping Offers New Paths to Health and Wellbeing

Why It Matters

Linked to better mental health, wellbeing, longevity, and resilience

Potentially modifiable: RCTs show gains in emotional regulation and compassion

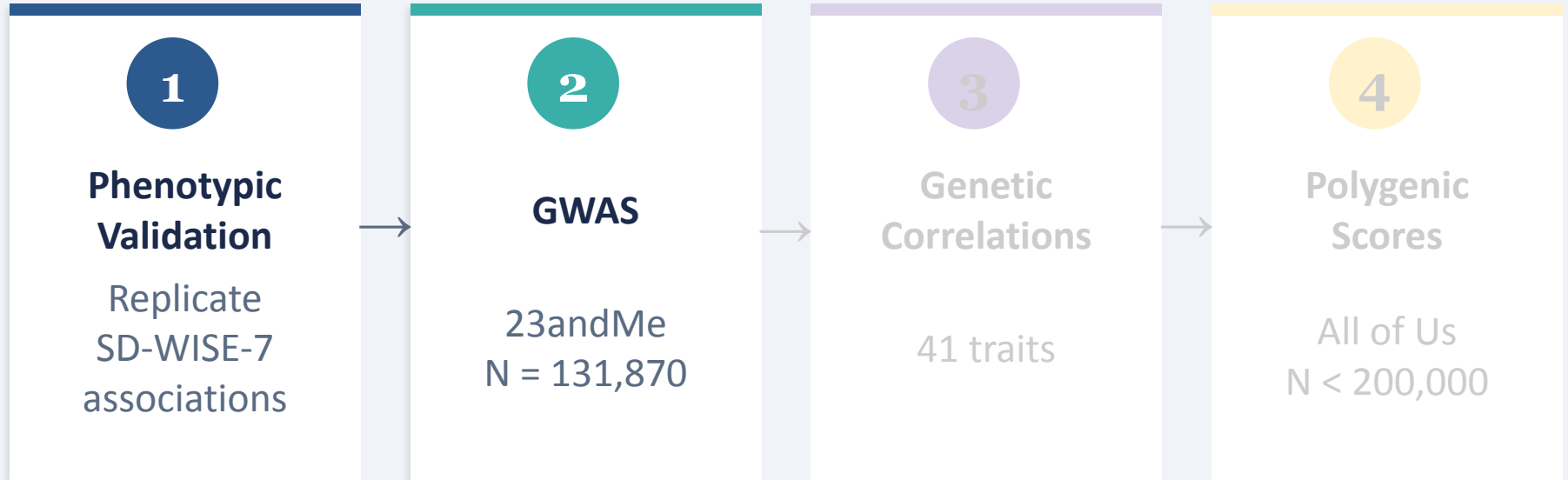
Potential application to therapeutics that improve health outcomes

Why We Expect a Genetic Basis

Grandmother Hypothesis: post-reproductive adults under selection for social advising and emotional regulation

Yet the genetic architecture of wisdom remains unmapped

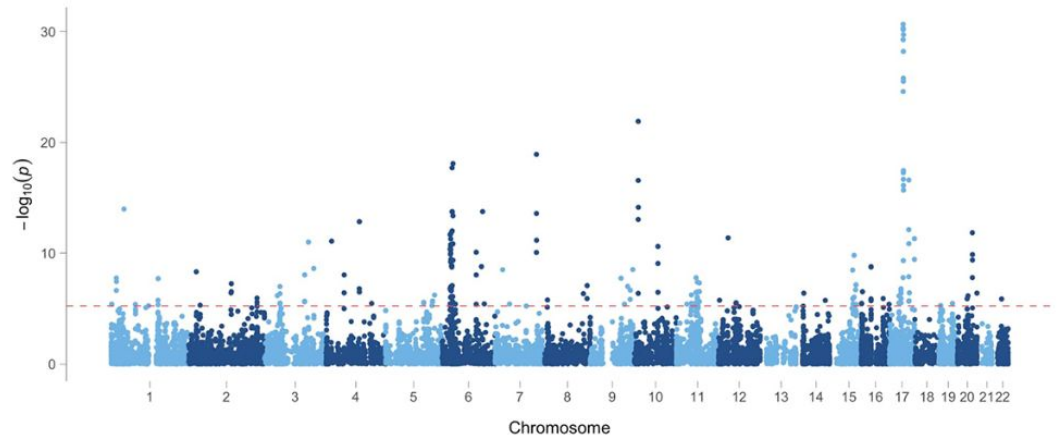
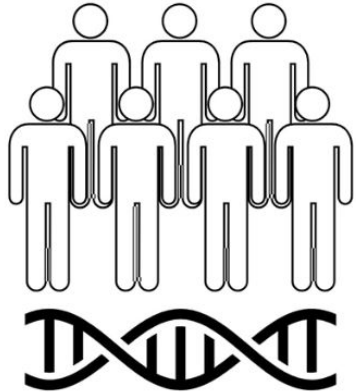
A Four-Stage Pipeline to Map Wisdom Genetics



Goal: Characterize the genetic architecture of wisdom and its relationship to health

Genome-Wide Association Studies (GWAS)

- **Goal:** Identifying genetic variants linked to specific traits
- **The Process:** Comparing the DNA of large groups of people to find variations that occur more frequently in those with the condition
- **The Result:** A "Manhattan Plot" that highlights which variations are most associated.



GWAS Results (23andMe, N = 131,870)

Heritability

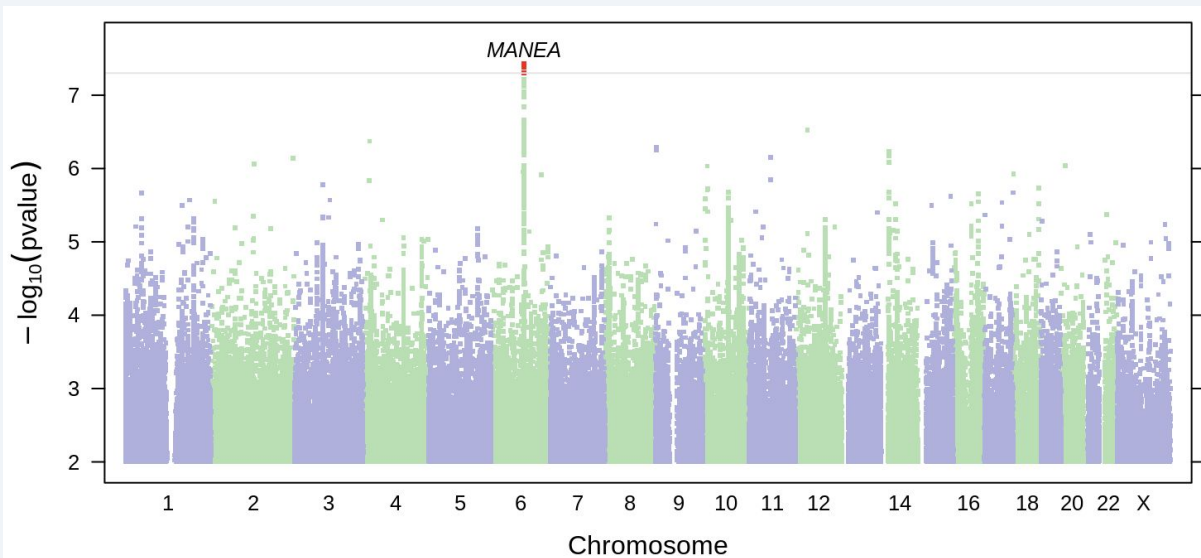
$h^2_{\text{SNP}} = 0.03$

SE = 0.004

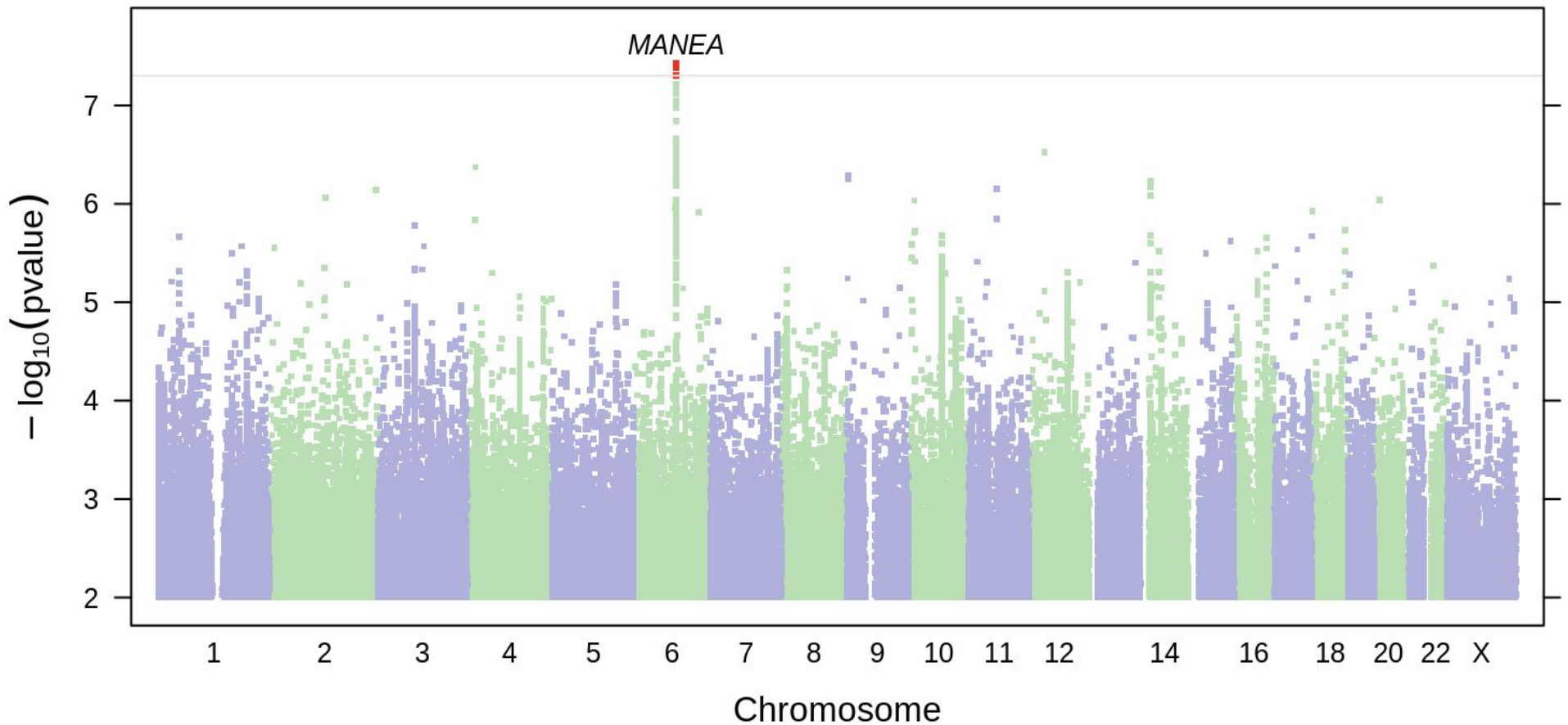
GW-Significant Locus

1 locus

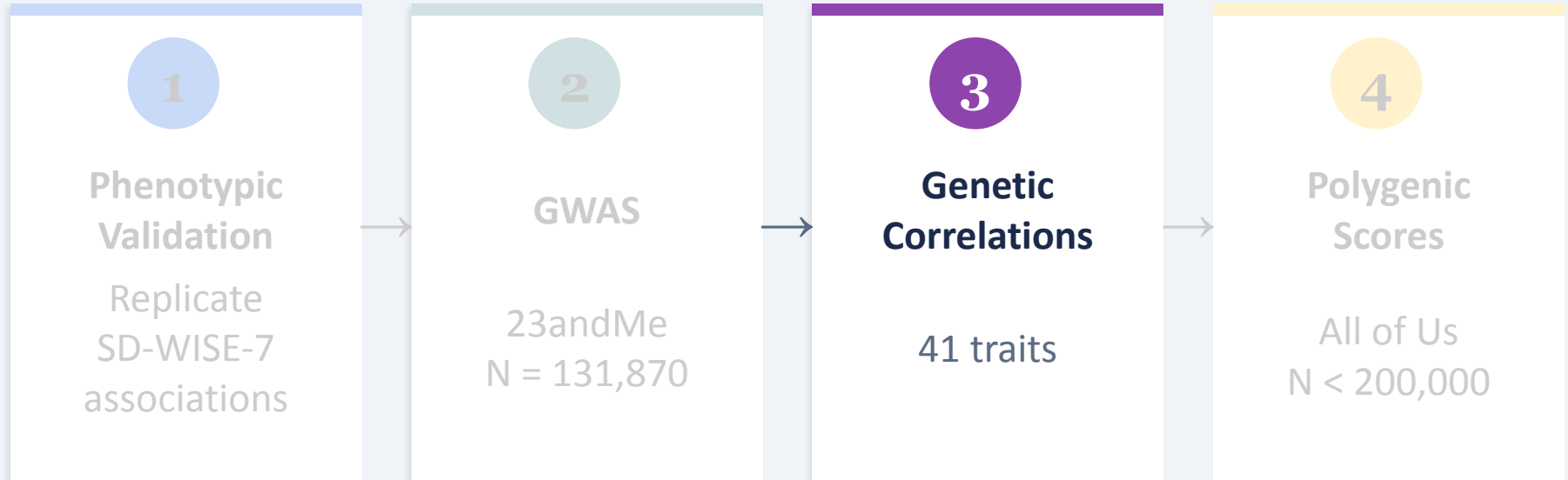
rs1028640 (chr6, MANEA)



GWAS Results (23andMe, N = 131,870)

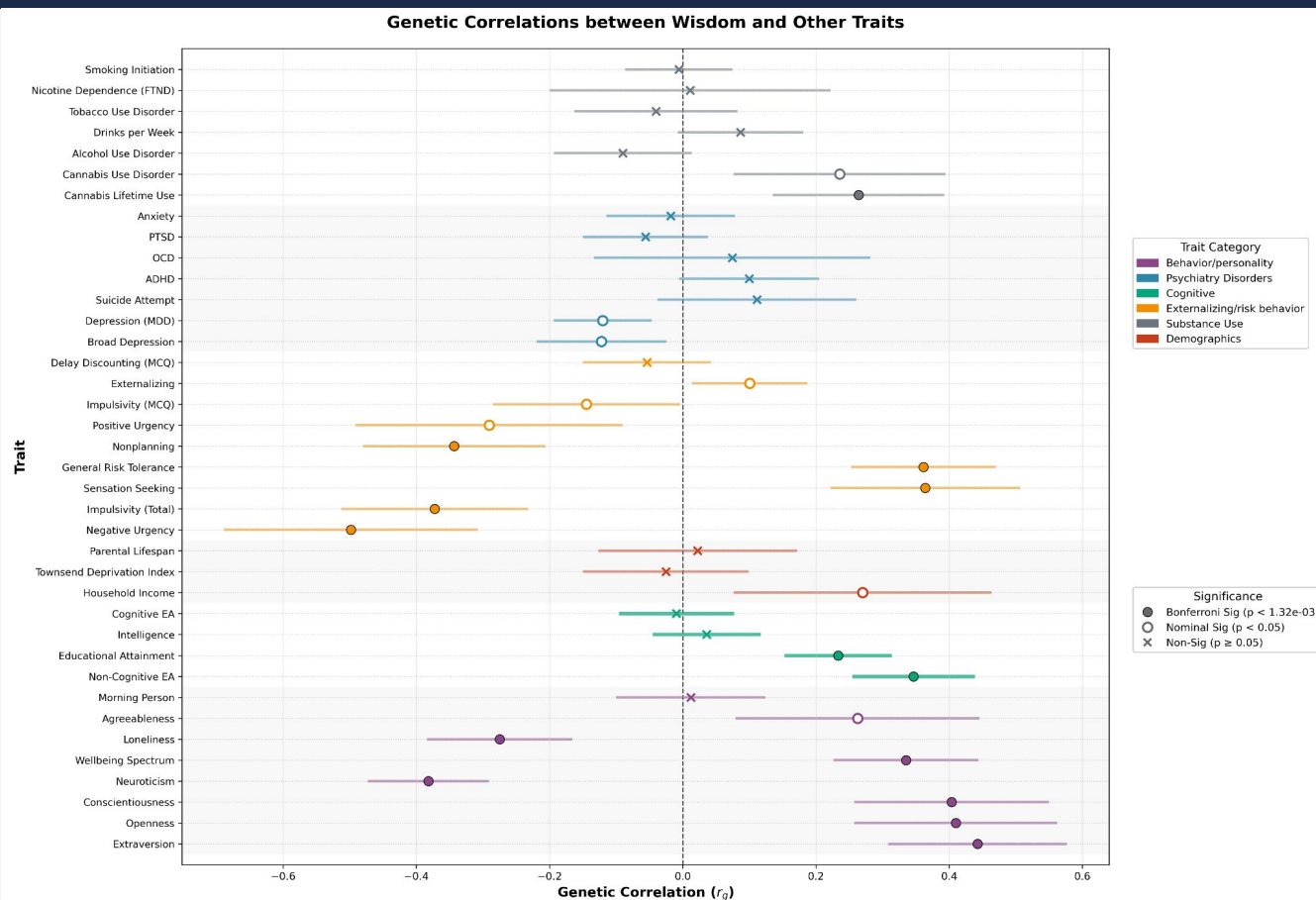


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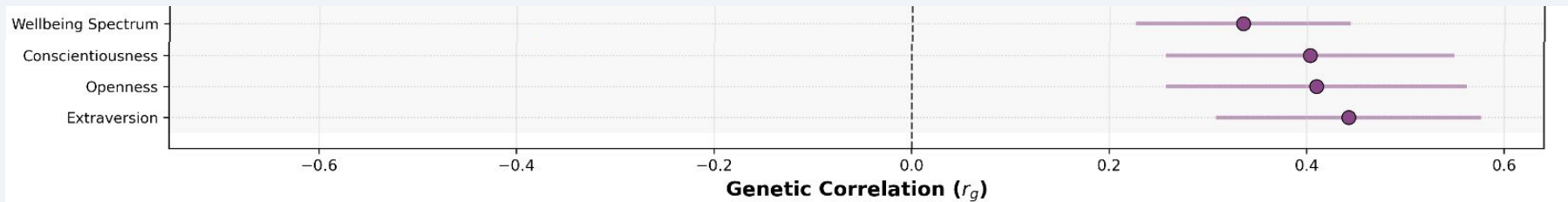


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Genetic Overlap Reveals Links to Mental Health and Personality



Wisdom Shares Genetics with Personality



Wellbeing

$r_g = 0.34$

Conscientiousness

$r_g = 0.40$

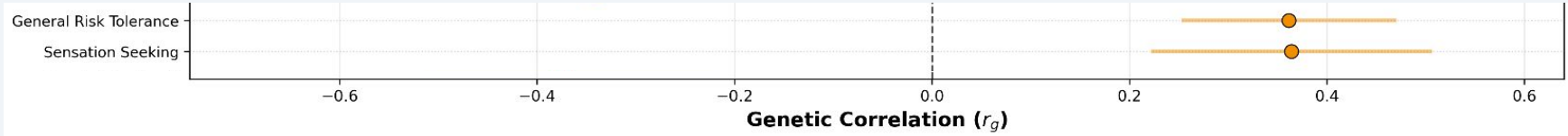
Openness

$r_g = 0.41$

Extraversion

$r_g = 0.44$

Wisdom Shares Genetics with Risk-Taking

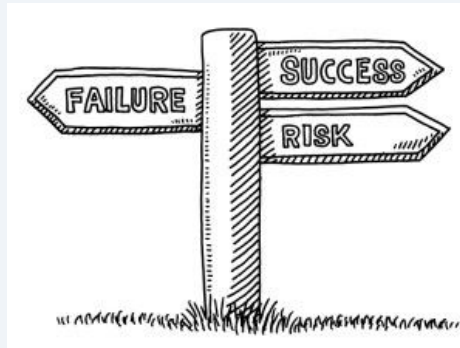


Risk Tolerance

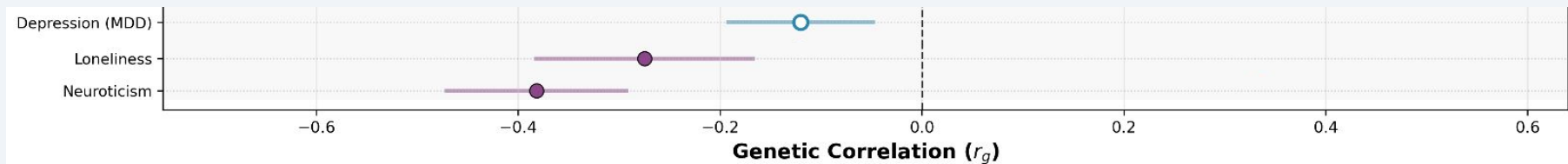
$r_g = 0.36$

Sensation Seeking

$r_g = 0.36$



Wisdom Diverges from Genetics of Adverse Traits



Depression (MDD)

$r_g = -0.12$

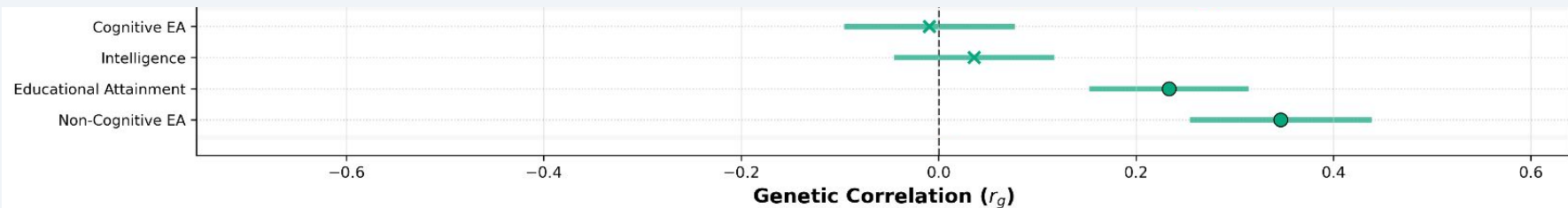
Loneliness

$r_g = -0.27$

Neuroticism

$r_g = -0.38$

Wisdom Is Character, Not IQ



Cognitive EA

$r_g = -0.01$ (ns)

Intelligence

$r_g = 0.04$ (ns)

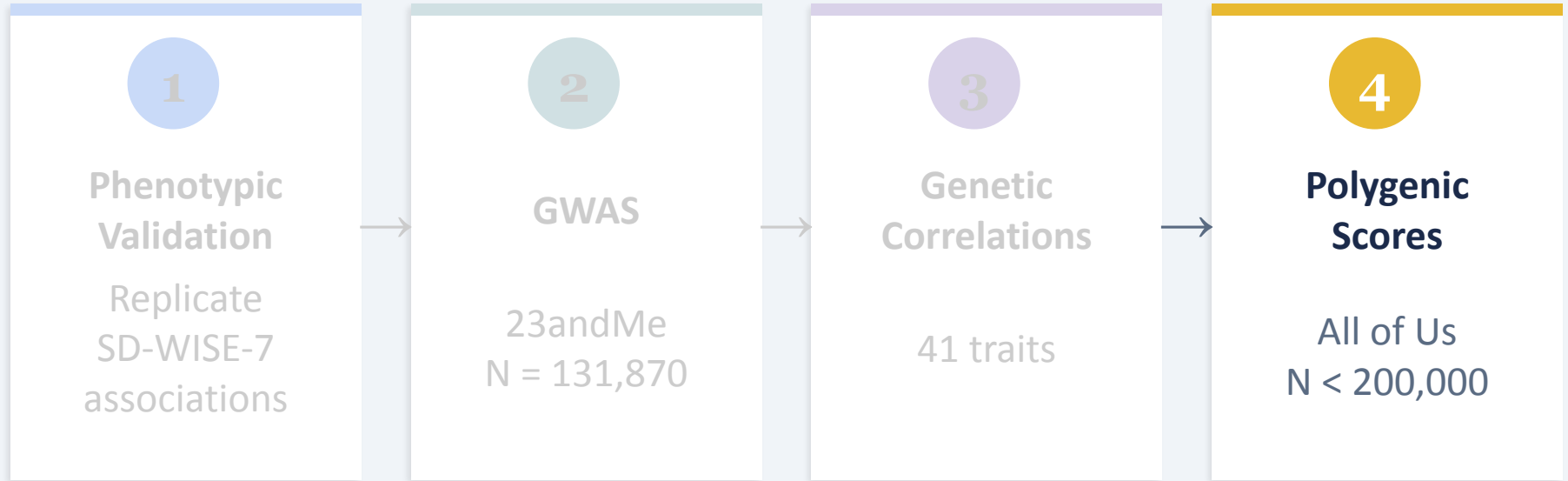
Educational Attainment

$r_g = 0.23$

Non-Cognitive EA

$r_g = 0.35$

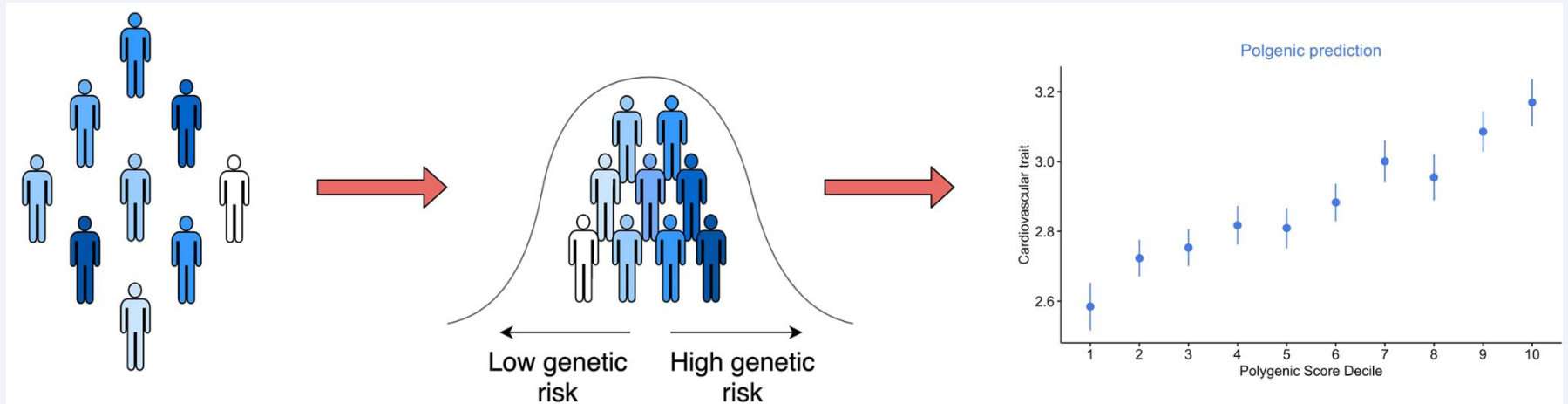
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Polygenic Scores (PGS)

- **Sum:** Aggregating thousands of small genetic signals from across the genome.
- **Rank:** Placing individuals on a distribution from lowest to highest score.
- **Relate:** Testing how the score connects to health and life outcomes.

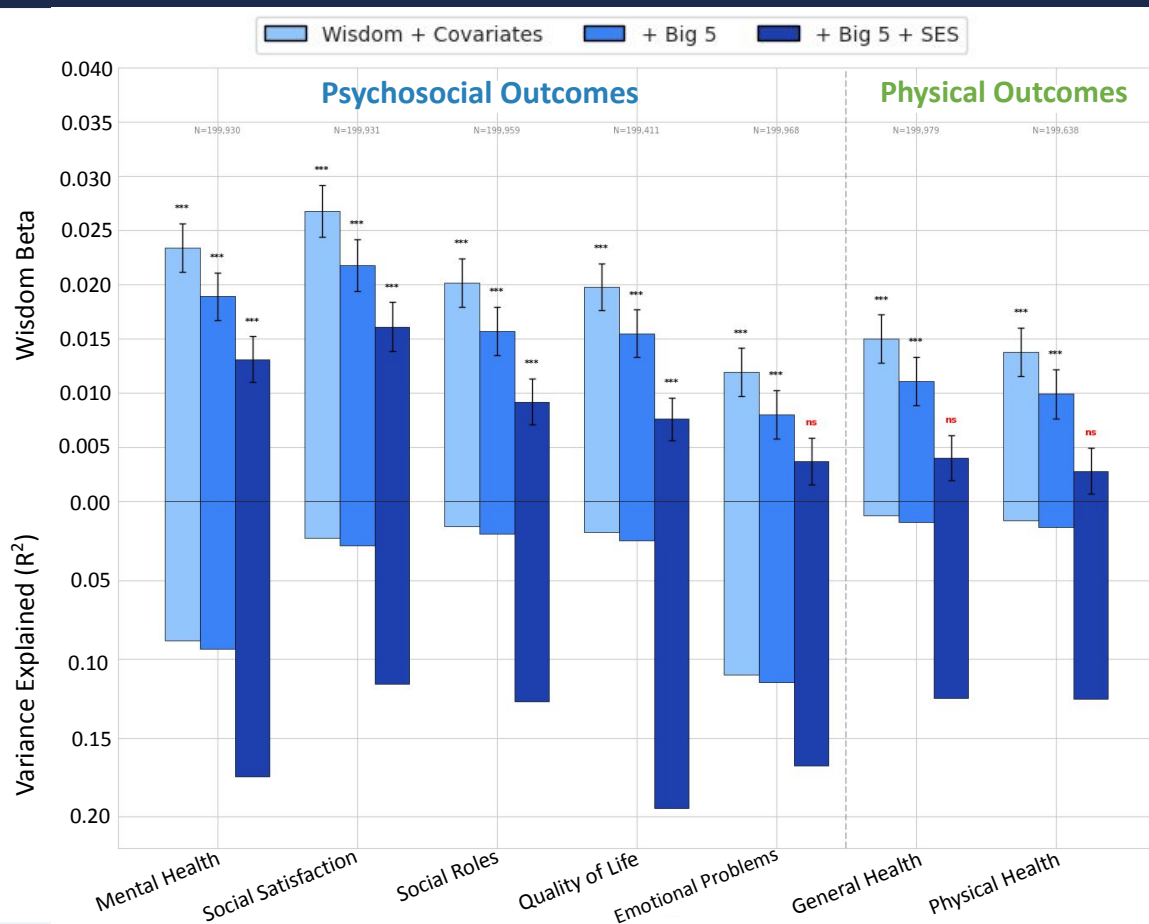


Wisdom PGS Validates Real-World Life Outcomes

All of Us | N < 200,000

Social Satisfaction	$\beta = 0.016$	$P = 1.2 \times 10^{-12}$
Mental Health	$\beta = 0.013$	$P = 5.9 \times 10^{-10}$
Social Roles	$\beta = 0.009$	$P = 1.4 \times 10^{-5}$
Quality of Life	$\beta = 0.008$	$P = 1.3 \times 10^{-4}$

PGS: Effect Sizes Across Health Outcomes



("Big 5" = Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism)

Conclusions

1

Wisdom is heritable and genetically mappable

$h^2_{\text{SNP}} = 0.03$ • One GWS locus near MANEA (chr6)

2

Genetically distinct from intelligence

No significant genetic correlation with Intelligence ($r_g = 0.04$) or cognitive educational attainment ($r_g = -0.01$)

3

Wisdom opposes psychological distress

Significant negative genetic correlations with neuroticism, loneliness, and depression

4

PGS predicts real-world outcomes

Wisdom PGS significantly predict social satisfaction, mental health, and quality of life in ~200K independent All of Us participants

Thank You!

Questions?

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